



SUSANTA HAZRA

AI Engineer • Machine Learning • LLM Applications • MLOps

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European Citizen

SUMMARY

AI Engineer with hands-on experience building and deploying end-to-end machine learning and LLM-powered applications. Skilled in Python, FastAPI, Docker, GCP, and MLOps workflows, with practical experience developing Transformer-based forecasting systems, NLP pipelines, and cloud-deployed AI services.

Recently completed the Le Wagon Data Science & AI Bootcamp and developed production-oriented AI projects involving time-series forecasting, REST APIs, automated data pipelines, and scalable deployment. Strong analytical and systems-thinking background from 12 years of scientific research and cross-disciplinary problem solving.

TECHNICAL SKILLS

Programming & Backend:

Python, SQL, FastAPI, REST APIs,

AI / Machine Learning:

Scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, Transformers

LLM & GenAI:

LangChain, Prompt Engineering, RAG Pipelines, AI Agents, Hugging Face

MLOps & Deployment:

Docker, MLflow, GCP, Streamlit, Git, BigQuery

Data Engineering & Visualization:

Pandas, NumPy, Matplotlib, Plotly, APIs, ETL Pipelines

AI Engineering Concepts:

Inference APIs, Cloud Deployment, Containerization, Time-Series Forecasting, Transformer Models, Pipeline Automation, Scalable ML Services

LANGUAGES

English Proficient ●●●●●

Portuguese Advanced ●●●●●

DEEPLARNING.AI CERTIFICATIONS

- LangChain for LLM Application Development
- Agentic AI
- ChatGPT Prompt Engineering for Developers
- AI Prompting for Everyone

PROJECTS



- Built a local multi-model AI platform using open-source LLMs (Mistral, Qwen, Gemma, Phi) with Ollama-based local inference.
- Designed a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB and sentence-transformers for semantic document retrieval.
- Engineered a modular FastAPI backend with LangGraph-based orchestration for multi-step AI workflows, model routing, and context-aware query execution.
- Architected a production-style LLM engineering system supporting local AI deployment, vector search, and extensible agentic workflows.

https://github.com/Susanta2025-lab/estudio-polymind-llm-orchestration

TF-IDF • NLP • Scikit-learn • FastAPI • Transformers

- Engineered an NLP machine learning pipeline utilizing custom text preprocessing and TF-IDF feature extraction (capturing 10,000 unigram/bigram features) to effectively handle high-dimensional sparse text matrices.
- Trained and hyperparameter-tuned linear and tree-based models (Logistic Regression, Random Forest, XGBoost) via RandomizedSearchCV, achieving a peak test accuracy of 97.08% and a cross-validation score of 0.9712 with an optimized XGBoost classifier.
- Architected a robust MLOps workflow by serializing the vectorizer and optimized estimators into unified Scikit-Learn Pipeline objects (.joblib), guaranteeing strict feature consistency between training and inference.
- Developed, containerized and deployed using FastAPI, Docker and Render to serve the optimized model, enabling seamless production deployment and real-time text classification over HTTP requests.

https://github.com/Susanta2025-lab/Fatocheck

PyTorch • FastAPI • Docker • GCP • BigQuery • Streamlit • MLflow • Time-Series Forecasting

- Designed and deployed an end-to-end AI forecasting platform for day-ahead electricity price prediction in the Germany-Luxembourg (DE-LU) energy market.
- Built and evaluated multiple forecasting models including Prophet, ARIMA, XGBoost, LSTM, and Transformer architectures to benchmark predictive performance.
- Developed a Transformer-based forecasting model in PyTorch and an XGBoost-based spike detection system for volatile energy price prediction.
- Built REST API services using FastAPI and containerized deployment workflows using Docker and Google Cloud Platform (GCP).
- Developed interactive visualization and forecasting interface using Streamlit for real-time prediction analysis.

https://github.com/xucenyng/grid-intelligence

EDUCATION



Data Science & AI Bootcamp

03/2026 - 05/2026

Le Wagon

- Built and deployed machine learning, deep learning, and LLM-powered applications using Python, FastAPI, Docker, and GCP.
- Developed time-series forecasting, NLP, and Transformer-based systems using Scikit-learn, TensorFlow/Keras, and PyTorch.
- Implemented MLOps workflows including MLflow, cloud deployment, API serving, and automated data pipelines.

EXPERIENCE



Assistant Researcher

2012 - 2024

Técnico Lisboa

Conducted advanced scientific research with strong emphasis on analytical modeling, experimentation, and data-driven problem solving